Product-only neural recommendations: training, evidence, and economics

Reproducible analysis of the supplied transaction CSV | 17 October 2026

Executive decision

A customer selects one product; the prototype offers exactly two complementary products for 15 seconds. Choosing either add-on in time applies a 15% discount to that add-on only. The historical data can support which products to test and the unit economics. It cannot establish that this promotion increases revenue or profit.

Recommendation: use the working local demo and run a randomized checkout pilot. Measure total basket gross profit and revenue per eligible checkout against a no-offer control. Do not describe the scenario figures in this report as a forecast.

Did the idea improve anything?

Historical prediction: modestly. On 6,186 May-June multi-product baskets, the selected neural network achieved 11.44% Hit@2 versus 10.96% for a product-pair rule and 9.98% for two popular items, a 1.46 percentage-point gain over popularity. This is retrospective prediction, not customer response to a prompt.

Sales impact: not measured. For illustration, assuming 5% genuinely new add-ons and 5% existing-buyer discount leakage, the six-month proxy gives $3,430 extra revenue, $1,461 extra gross profit, and $218 after assumed operating costs. With 10% leakage, that net becomes $-517. The CSV cannot establish either response rate; only a randomized pilot can establish uplift.

Evidence class	What is known
Observed	33,887 completed transactions, $305,925.14 recorded revenue.
Backtested	Historical top-two basket prediction on later months; not offer response.
Scenario	Financial effects under explicit assumed incremental and baseline-buyer rates.
Unknown	Actual offer exposure, incremental conversion, substitution, labor, inventory, and net profit.

Data source: stories_transactions.csv only. Transaction IDs begin with SYN-, a provenance signal that should be clarified before operational deployment. All final figures are regenerated by the repository code.

1. Data audit and scope

Measure	Result
Product lines	51,177
Transactions	33,887
Period	2026-01-01 to 2026-06-30
Branches / products / categories	3 / 45 / 9
Recorded revenue	$305,925.14
Recorded gross profit	$179,430.49
Missing customer ID rows	18,691
Transactions with a discounted line	3,247

Transaction ID defines the basket. Pair counts use distinct product presence, while financial totals preserve quantity and recorded line revenue/gross profit. Date, time, branch, order type, and customer ID are checked for within-transaction consistency. Source amounts reconcile within two cents per line; cent differences may reflect rounding. Customer ID is omitted from the recommender because of missingness and lack of a verified personalization benefit.

Offline pseudo-checkout: the file does not contain entry sequence. Each multi-product basket is evaluated from every possible seed item with total weight one per transaction. This estimates retrospective ranking quality, not an actual point-of-sale impression or response.

2. Preprocessing and numeric encoding

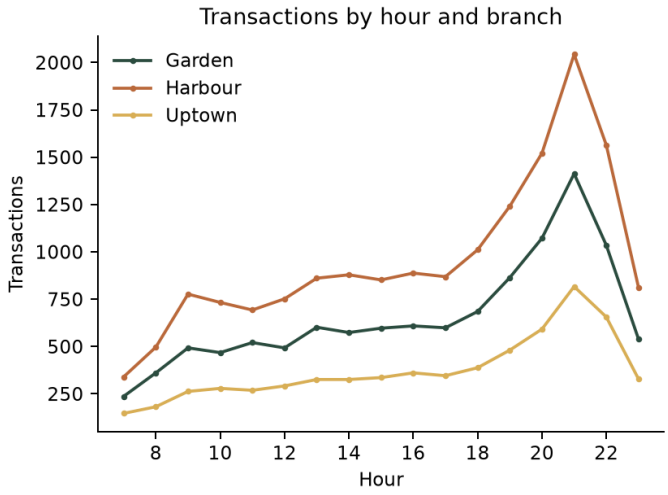
Every raw CSV field is converted to a numeric representation in an **audit-only** table. This makes missingness, type handling, and reproducibility explicit. The network is deliberately given only the selected product; transaction IDs, dates, branch, prices, discounts, and final revenue never enter its input.

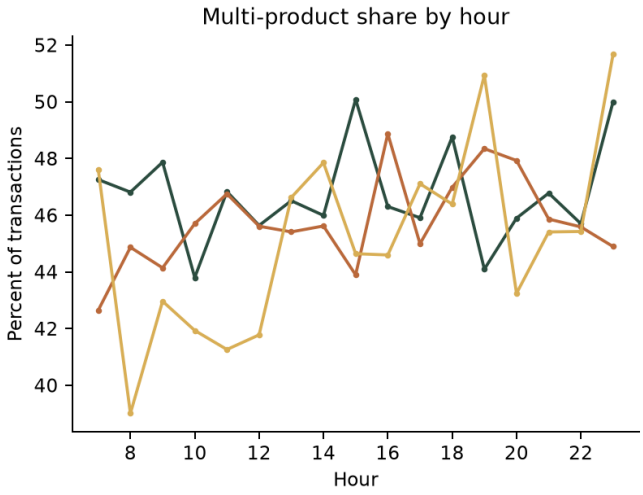
Field group	Audit/analysis transformation	Model input?
Transaction ID	Sorted integer code; grouping key	No
Date / Time	Days since 1970 / seconds after midnight	No - split and EDA only
Branch / Order Type / Category	Sorted integer code	No
Customer ID	Sorted integer code; missing = -1	No
Product	Sorted integer code; training-only vocabulary	Yes - one-hot
Quantity	Original positive integer	No
Unit Price / Unit Cost	Original USD numeric values	No - economics only
Discount Rate	Original decimal fraction	No - descriptive only
Line Revenue / Gross Profit	Original USD numeric values	No - outcomes only

Model normalization: a product is a 45-dimensional one-hot vector (one 1, all other entries 0); its Euclidean norm is already 1, so z-score scaling is unnecessary. Basket targets become a normalized distribution over the other distinct purchased products. Each basket contributes total training weight 1, regardless of whether it contains two or three products. Single-product baskets contain no positive complement label and are excluded from the ranking loss, but remain in EDA and opportunity sizing.

Leakage controls: vocabulary is learned on training records for evaluation; dates define chronological splits; no test basket updates the benchmark model. The final app model is retrained on all six months only after the test result is fixed. The numeric audit table is never passed to model.fit().

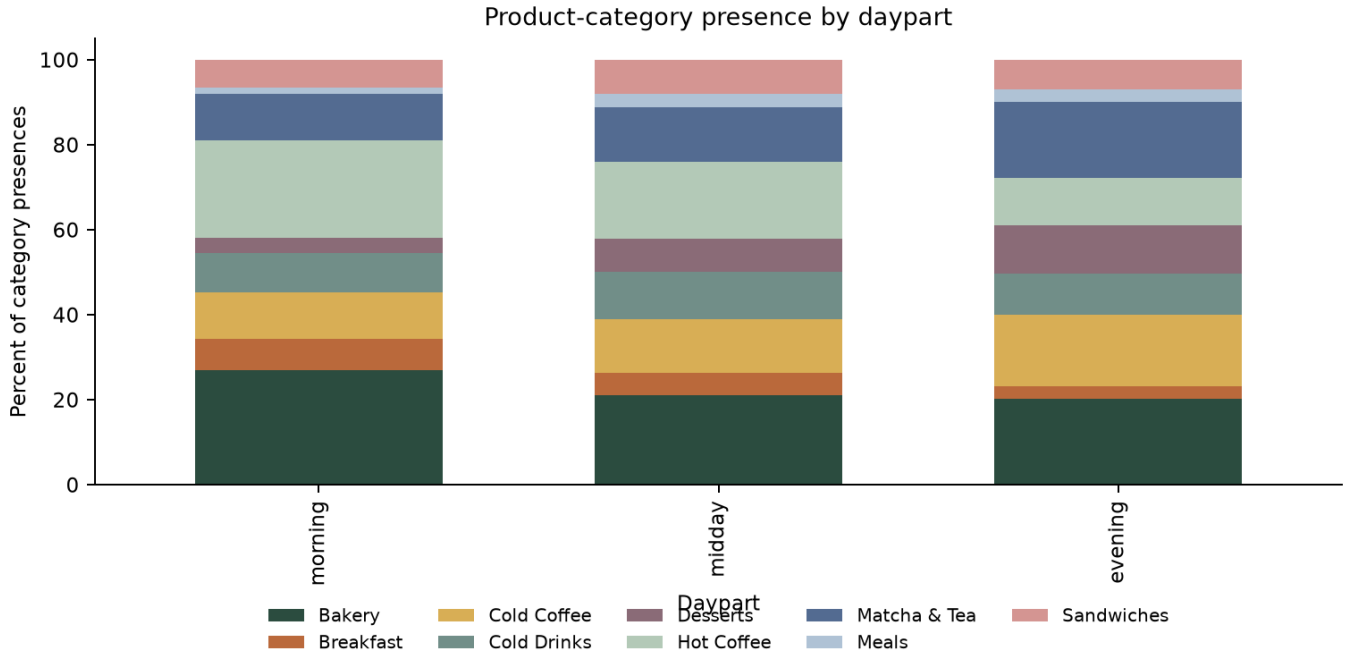
3. Time and branch





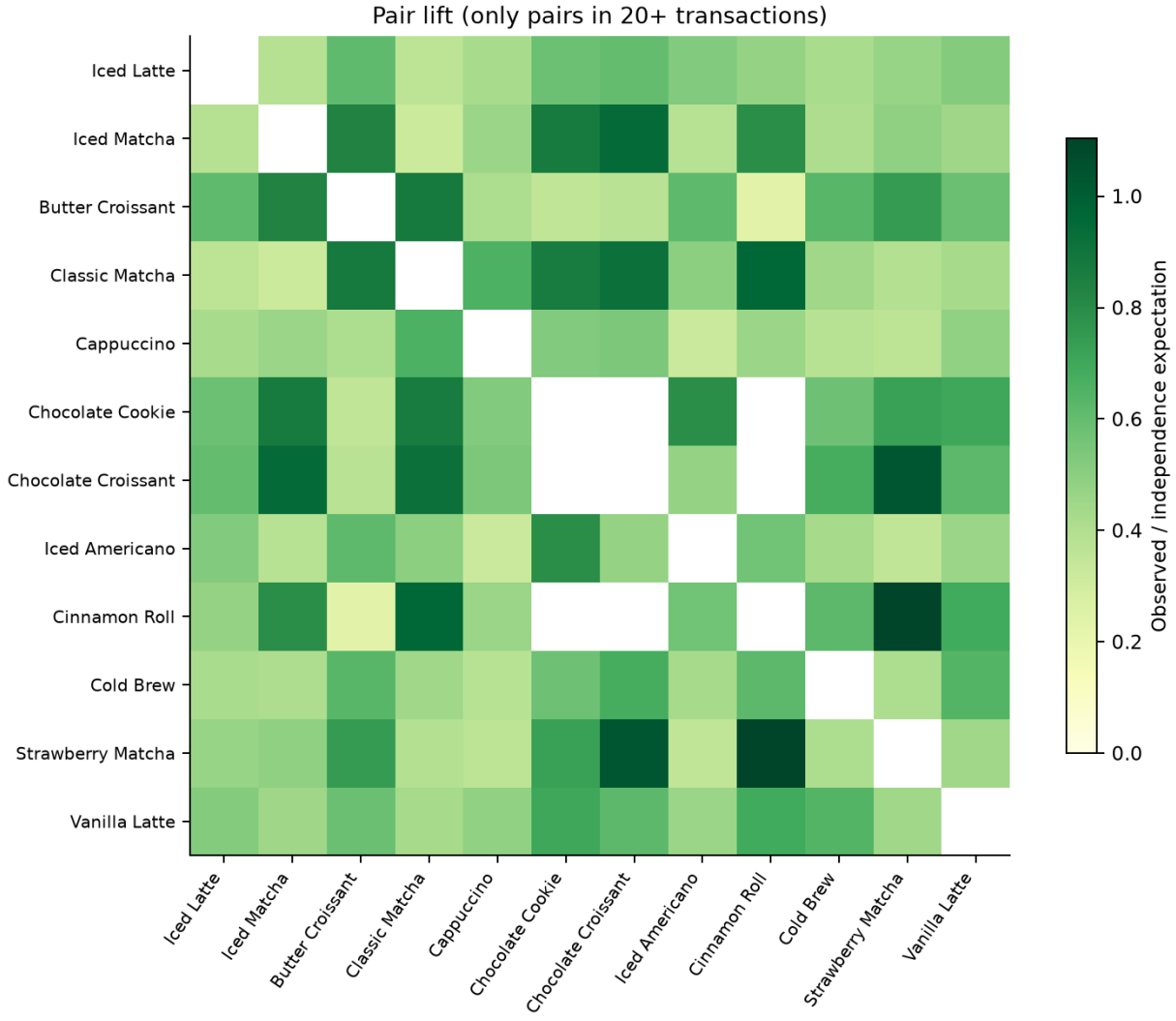
Counts use transactions per branch-hour. Multi-product share divides multi-product transactions by all transactions in each branch-hour. Traffic is not a measure of queue length or capacity.

4. Category mix by daypart



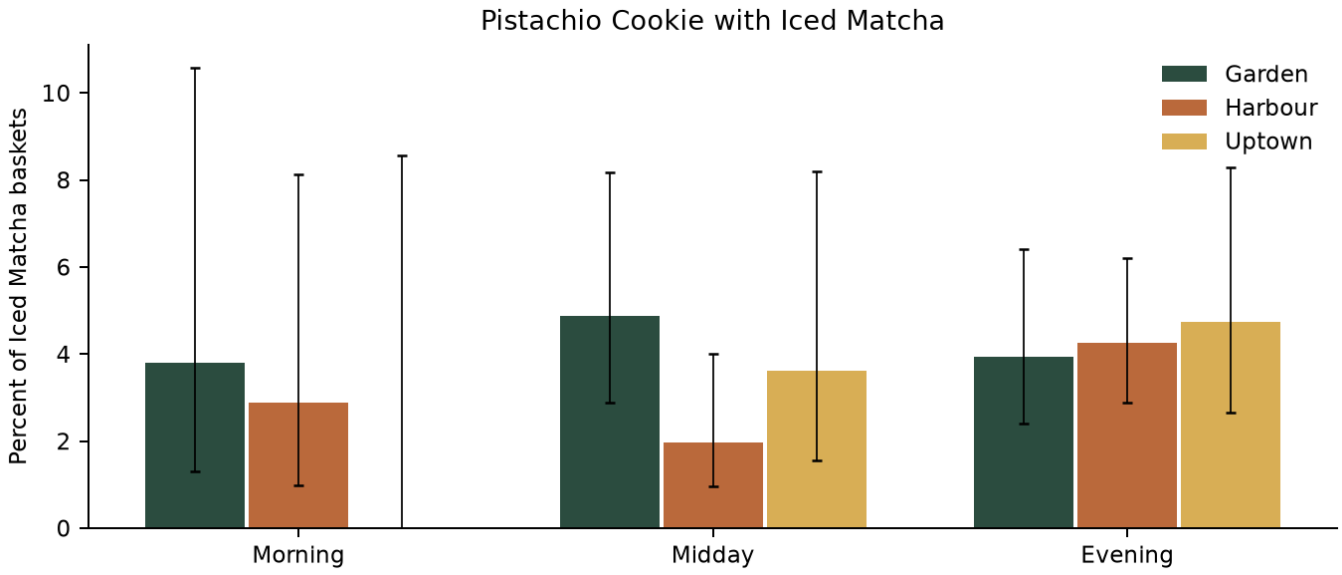
Each transaction/category presence is counted once. Shares are within daypart and describe product mix, not a probability that a recommendation will convert.

5. Product-pair association



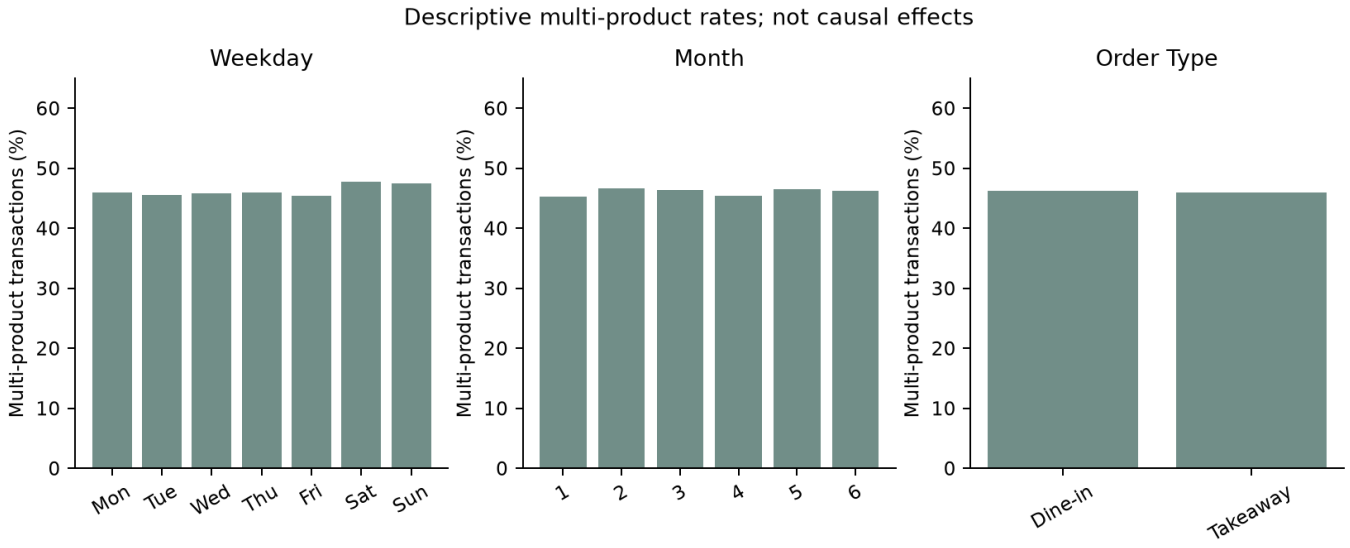
Lift compares observed pair frequency with the expectation under unconditioned independence. Only pairs with at least 20 transactions are shown; lift remains associational and can reflect context and basket size.

6. Example pairing



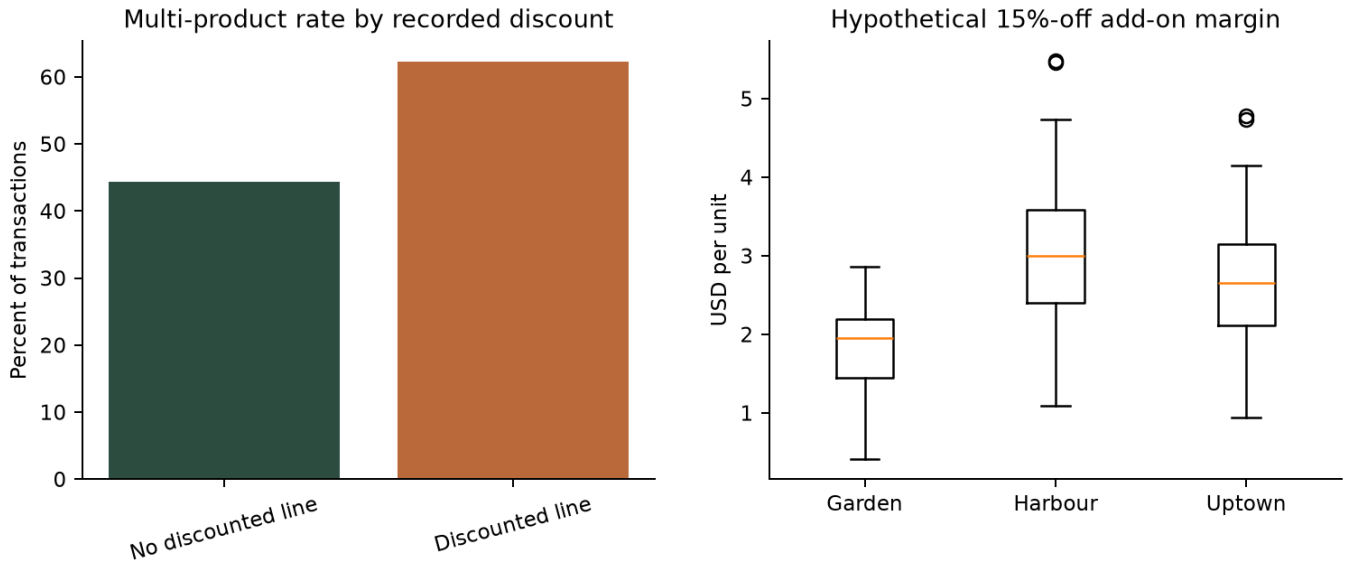
Iced Matcha and Pistachio Cookie occur together in 82 completed transactions. Bars divide pair baskets by seed-containing baskets in each branch and daypart; detailed cell counts are in outputs/results.json.

7. Other factors



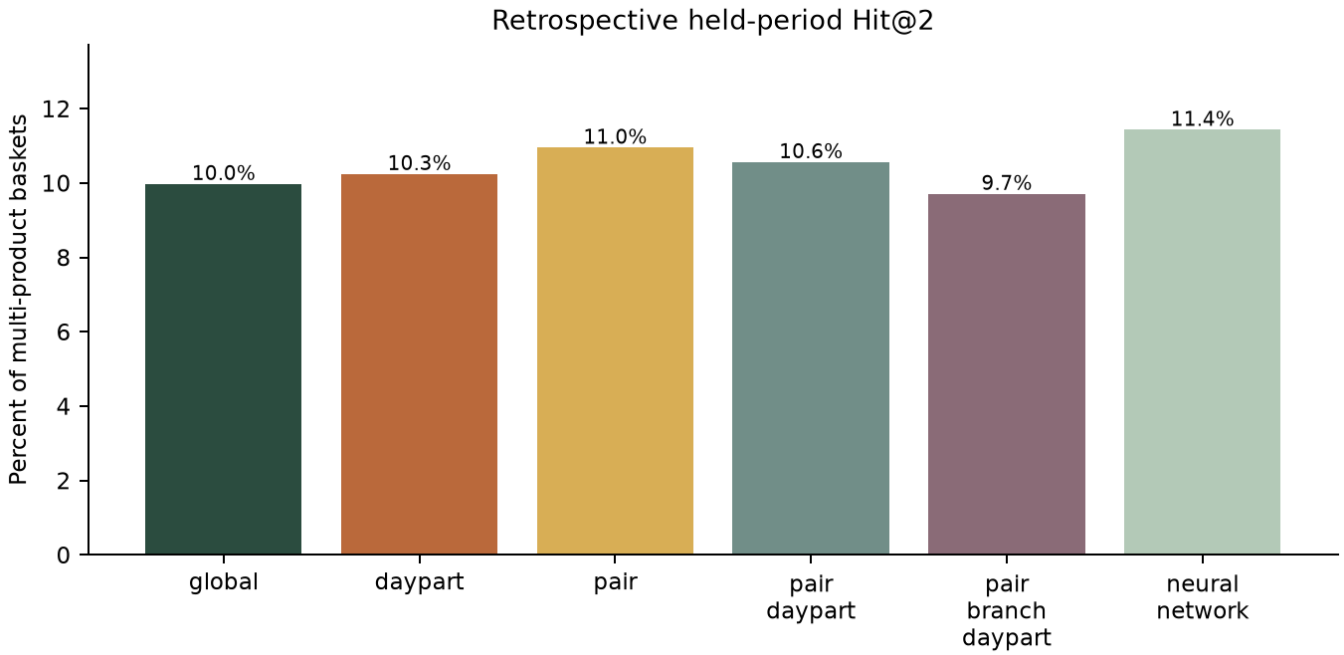
Weekday, month, and order type are plotted even when differences are small. Rates use all completed transactions in each group. Effects can be confounded by branch, product mix, and traffic.

8. Discount and margin



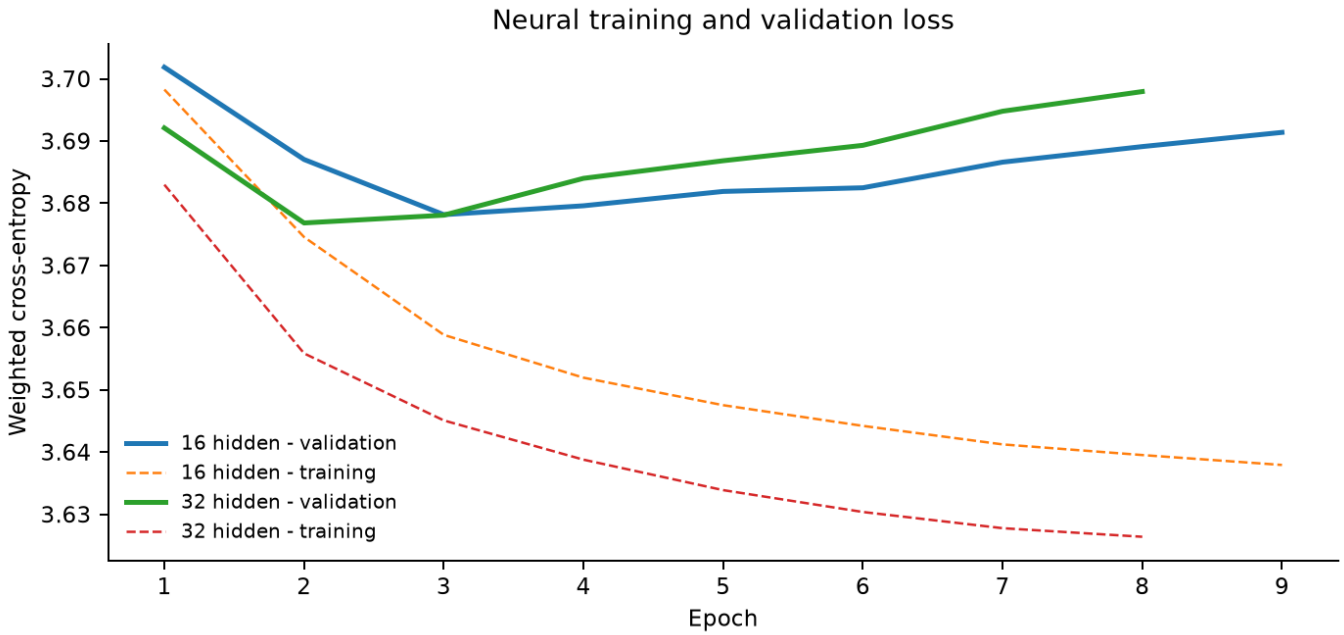
The left panel is a descriptive split by whether any line was discounted; it is not a causal discount estimate. The right panel uses branch-product median price and cost to calculate a hypothetical 15%-off unit margin.

9. Held-out model comparison



All models fit January-March and are evaluated on May-June after April validation selects neural architecture. The historical test period was discussed in previous project work, so this is retrospective evidence rather than a pristine prospective study.

10. Neural training decisions and learning curves



Two small networks were tried: 45 product inputs, either 16 or 32 tanh hidden units, and 45 softmax outputs. The output corresponding to the selected product is masked. Training uses transaction-weighted cross-entropy, Adam with learning rate 0.01, batches of 512, L2 weight decay 0.0001, a fixed seed, and early stopping after six epochs without lower April validation loss. The architecture is selected by April Hit@2, with validation loss as a tie-breaker; May-June is not used for tuning.

11. Train / validation / test results

Partition	Dates	Transactions	Multi-product baskets
Train	2026-01-01 to 2026-03-31	14,792	6,830
Validation	2026-04-01 to 2026-04-30	5,750	2,615
Test	2026-05-01 to 2026-06-30	13,345	6,186

Hidden units	Chosen epoch	April validation Hit@2	Validation loss
16	3	11.39%	3.678
32	2	11.31%	3.677

Selected network: 45 one-hot inputs -> 16 tanh hidden units -> 45 softmax outputs. The input is exactly one product; the final output is the two highest-scoring different products. The app loads weights refit on all six months for the preselected 3 epochs. Its held-out result below belongs to the pre-refit train/validation model, not the post-refit app weights.

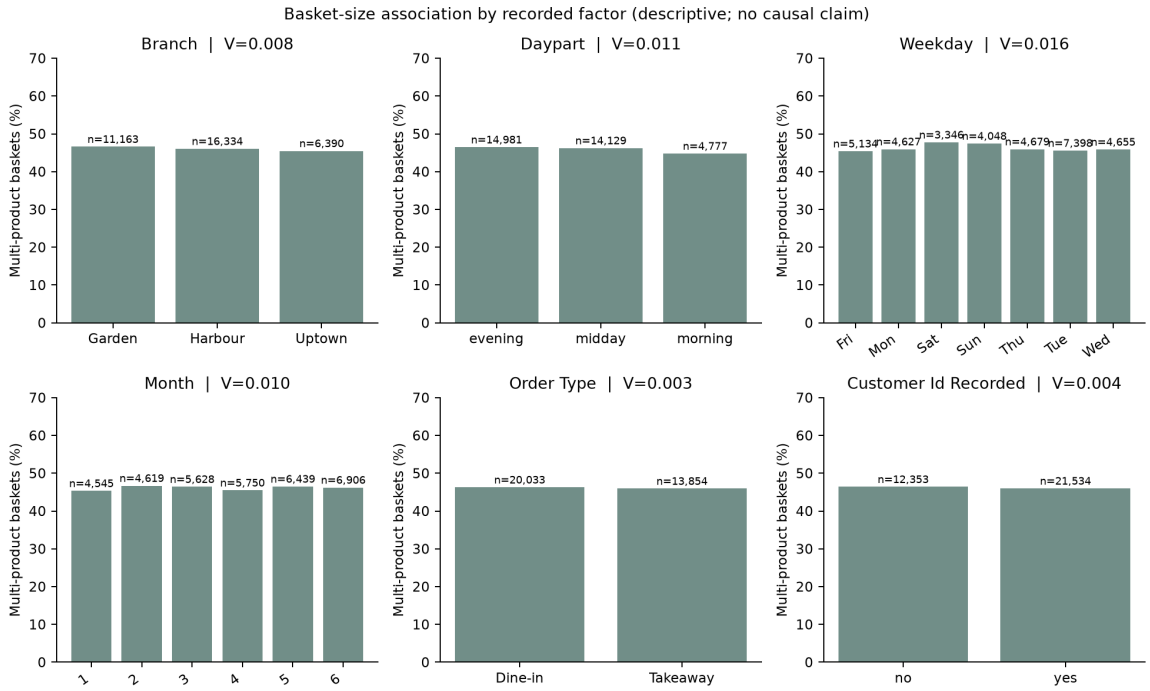
Rule	Hit@1	Hit@2 (95% CI)	Category Hit@2
global	5.24%	9.98% (9.43%-10.54%)	32.95%
pair	5.69%	10.96% (10.37%-11.55%)	33.56%
pair_daypart	5.72%	10.57% (9.97%-11.18%)	36.91%
Neural network	6.21%	11.44% (10.86%-12.03%)	32.31%

Hit@2 counts a transaction as successful when at least one of the two recommendations appears among its other purchased products. Every possible seed orientation is evaluated, weighted to total one per transaction. All shown models cover 100% of known seed products. MRR@2, branch/daypart breakdowns, full candidate histories, and exact values are in outputs/results.json. Intervals use transaction-level standard errors. This is basket prediction, not offer conversion or causal uplift.

Online learning: every confirmed multi-product demo order produces one gradient step at learning rate 0.025; an event ID prevents double updates. Single-product orders are logged but have no positive complement target. On restart, the app loads the saved six-month base weights and replays completed demo events once. No prompt impression is used as a training label.

12. Why product is the sole neural input: factor tests

We tested the claim that other CSV fields predict combo buying rather than assuming it. The outcome here is whether a completed transaction contains more than one distinct product. Cramer's V is an association effect size from 0 (none) to 1 (strong); these are descriptive relationships, not causal effects. A small p-value alone would be unhelpful at this sample size.

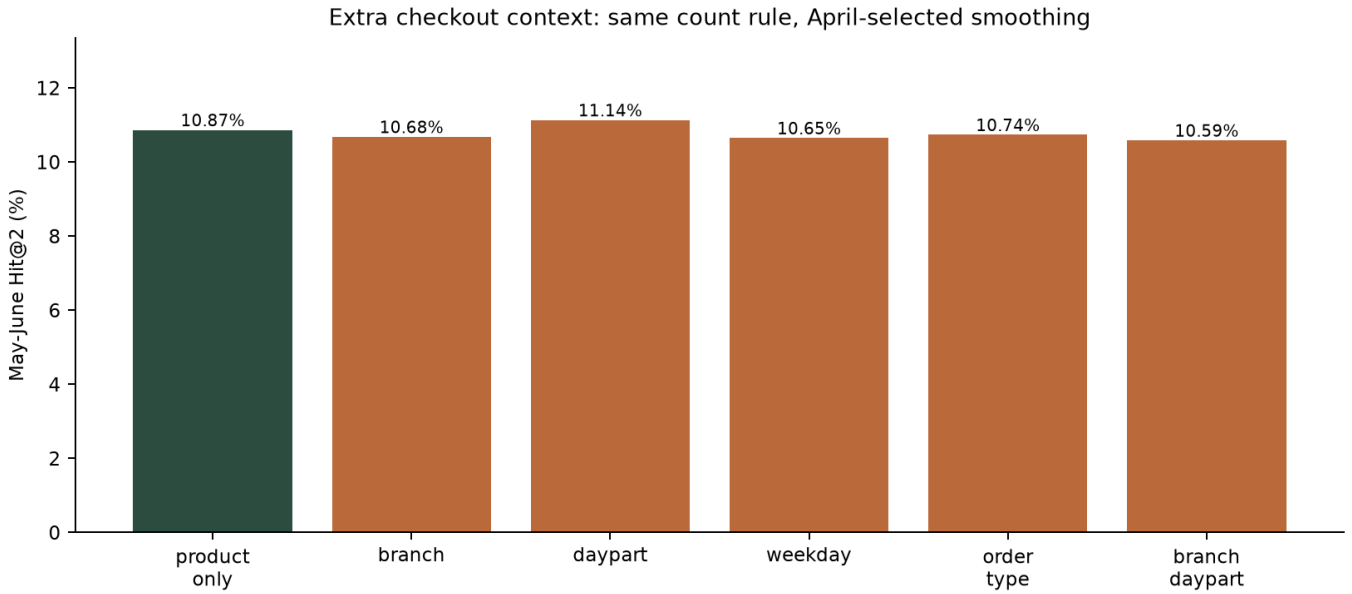


Recorded factor	Cramer's V	Lowest / highest multi-product rate	Use at offer time?
Branch	0.008	45.5% / 46.6%	Potentially
Daypart	0.011	44.8% / 46.5%	Potentially
Weekday	0.016	45.4% / 47.8%	Potentially
Month	0.010	45.3% / 46.7%	Potentially
Order type	0.003	45.9% / 46.3%	Potentially
Customer ID present	0.004	46.0% / 46.4%	Potentially
Discounted line	0.105	44.4% / 62.2%	No - post-purchase

Branch, daypart, weekday, month, order type, and recorded-customer availability all have V below 0.02. The discounted-line indicator has a larger association, but is determined during the completed sale; using it at first-product selection would leak the outcome. Customer ID itself is an identifier with substantial missingness, not a numeric quantity with meaningful distance.

13. Do extra checkout inputs improve recommendations?

We also tested whether known checkout context improves the **identity of the companion product**. This diagnostic keeps the same smoothed product-pair count rule and adds one context at a time. It fits January-March, picks smoothing on April, and reports May-June Hit@2. It is a feature ablation for a simple comparator, not a retraining of the neural architecture. Exact month is not tested as a context key because future months have no matching training observations.



Count-rule input	May-June Hit@2	Change vs product only (paired 95% CI)
Product only	10.87%	Reference
Product + branch	10.68%	-0.19 pp (-0.418, +0.046)
Product + daypart	11.14%	+0.27 pp (-0.003, +0.537)
Product + weekday	10.65%	-0.22 pp (-0.425, -0.011)
Product + order type	10.74%	-0.13 pp (-0.485, +0.221)
Product + branch + daypart	10.59%	-0.28 pp (-0.716, +0.151)

Daypart has the highest point estimate for this count rule, but its paired interval includes zero; the other tested contexts show no reliable held-out gain over the product-only version. Intervals account for pairing on the same test baskets, not for all model-selection uncertainty. The product-only neural network scored 11.44% Hit@2 on this period. We keep its input to the selected product because that is available at the exact offer moment, is simpler to update, and meets the product brief. These experiments do not prove that all contextual signals are useless; a prospective test or a controlled neural input ablation would be needed for that stronger claim.

14. Financial sensitivity - example, not forecast

Example: Harbour, first item Iced Matcha, offered add-on Pistachio Cookie. Median add-on price $3.62; median recorded unit cost $1.18. At 15% off, the rounded add-on selling price is $3.08 and unit gross profit before extra operating costs is $1.90.

The CSV contains 1,048 Harbour baskets with the seed and 375 completed seed-only baskets. Neither count is a known number of live offer impressions.

Per eligible offer, under the stated simplifying assumptions: incremental revenue = newly added rate x discounted price - rate of existing buyers receiving a discount x concession - substituted revenue. Incremental gross profit replaces price with discounted unit gross profit, then subtracts discount leakage, substitution, and added variable operating cost. All response rates are unknown until tested.

Assumed new add-on	Existing buyers discounted	Revenue / offer	Gross profit / offer
3%	0%	$0.092	$0.057
3%	5%	$0.065	$0.030
3%	10%	$0.038	$0.003
5%	0%	$0.154	$0.095
5%	5%	$0.127	$0.068
5%	10%	$0.100	$0.041
10%	0%	$0.308	$0.190
10%	5%	$0.281	$0.163
10%	10%	$0.254	$0.136

At an assumed 5% existing-buyer discount leakage and zero substitution/extra cost, new add-on conversion would need to exceed 1.4% for positive incremental gross profit in this simplified example. Larger operating or substitution costs raise this threshold.

15. Portfolio-scale sensitivity

There are 18,256 completed one-product transactions. Using them as a historical sizing proxy, the final neural model generates two offers for each basket. An equal mix of the two offers has mean discounted selling price $4.56, mean discounted unit gross profit $2.41, and mean discount concession $0.81. These item amounts are computed from the CSV; the 50/50 choice mix and response rates are assumptions. One-product completed receipts are not known live offer impressions.

The following six-month numbers multiply per-offer scenarios by that sizing proxy. They are **not forecasts**. Net scenarios additionally assume $0.25 per newly added item, $0.02 per offer, $25 monthly fixed expense, and $500 one-time development expense. No such operating cost is recorded in the CSV; replace each assumption with a quote or pilot observation before a rollout decision.

New add-on	Existing buyer discount	Extra revenue	Extra gross profit	Net after assumed costs
3%	0%	$2,499	$1,318	$166
3%	5%	$1,764	$583	$-569
3%	10%	$1,030	$-152	$-1,304
5%	0%	$4,165	$2,196	$953
5%	5%	$3,430	$1,461	$218
5%	10%	$2,696	$726	$-517
10%	0%	$8,330	$4,392	$2,920
10%	5%	$7,595	$3,657	$2,186
10%	10%	$6,861	$2,922	$1,451

These calculations assume no substitution and no effect on first-item sales. Either could make the outcome worse. Conversely, repeat visits or a different offer mix could change it. The randomized pilot must measure full baskets, not only recommended-item redemptions.

16. Marketing, operations, and pilot

Customer presentation: after the first item, show two relevant add-ons, each original and discounted price, and a plain 15-second timer. Discount only a selected add-on. Monitor whether urgency increases abandonment or dissatisfaction. Branch and daypart can be used for targeting only when retrospective evidence supports them.

Cost	CSV evidence	Needed before rollout
Discount concession	Listed price and 15% rule	Actual redemptions and baseline buyers
Product cost	Recorded unit cost by branch	Confirm cost definition
Development / hosting	None	Time log and supplier quote
Staff / prep / packaging	None	Observed incremental minutes and costs
Stockouts / substitution	None	Inventory and full-basket outcomes

Minimum randomized pilot: assign eligible checkouts before the offer is displayed to either ordinary checkout or the exact two-item, 15-second, 15%-off experience. Keep assignment stable on refresh and log both shown items, expiry, response, completed basket, revenue, gross profit, and order completion. Analyze all assigned checkouts, including refusals/timeouts. Primary endpoint is total basket gross profit per eligible checkout with a confidence interval; revenue per eligible checkout is a co-primary business view or key secondary metric. Obtain pilot variance before final sample sizing.

If volume permits, add an otherwise identical no-discount arm to separate suggestion effects from the discount. Measure baseline-buyer discount leakage and substitution across the whole basket. Scale only when incremental contribution exceeds operating costs without unacceptable customer or service effects.

Reproduction: run `python -m src.pipeline` from the repository root, then `streamlit run app.py`. The CSV remains the only source of historical business figures; outputs/results.json contains the complete computed tables and cell counts.